Ramón Darío Iglesias

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Recent Experience

Clementine, San Francisco, CA **09/2023 – Present**
Founder / CEO
Founded an AI x games company and led development of agentic systems, LLM infrastructure, and evaluation frameworks:

- Led the design and development of *Allyon: Battlecry*, a monster-trainer game where players train monsters using voice;
  - Implemented the voice-to-code pipeline that generates, compiles, and executes C# behaviors in real time.
  - Implemented safety guardrails in-game by sandboxing what the LLM can access and filtering disallowed outputs (e.g., emoji-based feedback channels with blacklisted symbols).
- Led the design and development of clm, a framework to integrate AI models in games
  - Developed a multimodal AI gateway that routes and serves different LLMs and multimodal models (chat, images, audio, streaming) behind a unified interface with observability.
  - Designed and implemented an SDK for integrating generative models into the Godot engine, including image, audio and text models, local RAG, object hydration, and bindings for C# and GDScript.
  - Built observability and evaluation infrastructure for the SDK and AI gateway: telemetry for prompts/tool calls, automated benchmarking for code generation, and metrics on compile success, retries, and latency. These metrics helped us polish the API we expose to the LLM leading us to reduce code generation failure rate by 66% and latency by 75%.
- Built *Thus Spoke Zaranova*, an AI social deception game where NPCs are LLM agents using RAG, reflection, and tool-calling to infer hidden roles and adapt strategies in multiplayer sessions.
- Prototyped colony-sim and god-sim games where pawns are LLM-driven agents combined with symbolic planners.
- Built and led a cross-functional team of AI engineers, gameplay engineers, technical designers, and artists.
- Raised $750k from a16z Speedrun and angel investors.
- Created the AIxGames community (meetups, Discord, *Playing With Inference* podcast) to share best practices in LLM agents, tools, and game applications.

Lyft Inc., San Francisco, CA **09/2019 – 09/2023**
Senior Data Scientist
Tech lead building real-time optimization and ML systems for fleet management:

- Increased revenue 5% in selected markets by designing, developing and deploying Priority Mode Controller, a system to decide in real time how many drivers should we give matching preference to [1].
- Built driver repositioning system optimizing earnings via spatiotemporal demand forecasts, effectively treating drivers as agents in a large-scale stochastic control problem.
- Led A/B testing framework evaluating driver nudges, influencing real-time supply management strategy.
- Reduced driver's phone pose estimation compute time by ∼90% via an eigenvector-based reformulation.
- Doubled stop sign detection coverage using computer vision and driver telemetry.
- Built ML workflow to detect road obstacles (e.g. potholes, speedbumps) based on phone IMUs' data.

Stanford University, Stanford, CA **01/2016 – 08/2019**
Research Assistant
Developed methods for large-scale autonomous fleet coordination:

- Built queueing-theoretical framework for heterogeneous fleet constraints, cast as linear program for real-time optimization [7, 5].
- Developed stochastic MPC using RNN demand forecasts to operate autonomous fleets, achieving 90% reduction in wait times [6, 2].
- Deployed vehicle relocation system with Toyota Ha:mo (Japan): 5% increased availability, 10% parking improvement.
- Side projects: multi-task learning's impact on CNN representations; deep learning for Bayesian filters from images.

Lyft Inc., San Francisco, CA **06/2018 – 09/2018**
Research Science Intern

- Developed stochastic programming algorithm for real-time autonomous fleet repositioning using demand forecasts. Deployed to Las Vegas self-driving pilot.
- Performed steady-state analyses (network flows, queueing theory) and power analyses to design production experiments.

SunPower, Richmond, CA **01/2015 – 01/2016**
Software Engineer

- Built pricing engine powering $5M+ yearly in residential leases. Developed data processing web app and CI/CD pipeline.

Education

Stanford University, Ph.D. in Civil Engineering **01/2016 – 08/2019**
Focus: Optimal Control of Autonomous Fleets.
Advisor: Marco Pavone

Stanford University, M.S. in Civil Engineering **09/2012 – 06/2014**
Focus: Data Analysis, Optimization, Machine Learning

UT Austin, B.S. in Civil Engineering **08/2007 – 06/2012**

Skills

LLMs & Agents: RAG systems, agent architectures, observability, evaluations;
ML/AI: PyTorch, TensorFlow, Burn (Rust ML);
Languages: Rust, Python, TypeScript/JavaScript, C#, GDScript
Infrastructure: AWS, Docker, PostgreSQL, Git, OpenTelemetry;
Methods: Optimization (LP, MPC), RL, control theory, queueing theory, A/B testing

Publications

[1] V. Krishnan, R. Iglesias, S. Martin, V. Pattabhiraman, S. Wang, and G. van Ryzin. Solving the ride-sharing productivity paradox: Priority dispatch and optimal priority sets. *INFORMS Journal on Applied Analytics*, 2022. INFORMS Wagner Prize Finalist, 2021.

[2] M. Tsao, R. Iglesias, and M. Pavone. Stochastic model predictive control for Autonomous Mobility on Demand. In *Proc. IEEE Int. Conf. on Intelligent Transportation Systems*, 2018.

[3] F. Rossi, R. Iglesias, M. Alizadeh, and M. Pavone. On the interaction between Autonomous Mobility-on-Demand systems and the power network: Models and coordination algorithms. *IEEE Transactions on Control of Network Systems*, 2018.

[4] F. Rossi, R. Iglesias, M. Alizadeh, and M. Pavone. On the interaction between Autonomous Mobility-on-Demand systems and the power network: Models and coordination algorithms. In *Robotics: Science and Systems*, 2018.

[5] R. Iglesias, F. Rossi, R. Zhang, and M. Pavone. A BCMP network approach to modeling and controlling autonomous mobility-on-demand systems. *Int. Journal of Robotics Research*, 2018.

[6] R. Iglesias, F. Rossi, K. Wang, D. Hallac, J. Leskovec, and M. Pavone. Data-driven model predictive control of Autonomous Mobility-on-Demand Systems. In *Proc. IEEE Conf. on Robotics and Automation*, 2018.

[7] R. Iglesias, F. Rossi, R. Zhang, and M. Pavone. A BCMP network approach to modeling and controlling Autonomous Mobility-on-Demand systems. In *Workshop on Algorithmic Foundations of Robotics*, 2016.